

Lesson 08: Linear Second-Order ODEs - Non-Homogeneous Case

Professor Dave Explains Differential Equations Master Series · Lesson 8

Video: Linear Second-Order Differential Equations Part 2: Non-Homogeneous Differential Equations

Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that non-homogeneous ODEs $ay'' + by' + cy = g(x)$ represent physical systems subjected to external forces—like a suspension driven by road bumps, or a circuit driven by an AC voltage. By the Superposition Principle, the general solution is the sum of the system's natural free response $y_c(x)$ and its forced response $y_p(x)$: $y(x) = y_c(x) + y_p(x)$. We use Undetermined Coefficients for standard polynomial/exponential/sinusoidal drives, and Variation of Parameters for arbitrary functions like $\ln x$ or $\sec x$.

2. Core Theoretical Framework & Rigorous Formulations

1. Undetermined Coefficients Guess Table & Resonance Rule

Driving Function $g(x)$	Initial Trial Guess $y_p(x)$
$P_n(x) = a_nx^n + \dots + a_0$	$A_nx^n + \dots + A_1x + A_0$
e^{kx}	Ae^{kx}
$\cos(kx)$ or $\sin(kx)$	$A\cos(kx) + B\sin(kx)$
$e^{\alpha x} \cos(\beta x)$	$e^{\alpha x} (A\cos(\beta x) + B\sin(\beta x))$

The Multiplication (Resonance) Rule: If any term in the trial guess duplicates a solution in $y_c(x)$, multiply the entire trial guess by x (or x^2 for double roots) until all duplication disappears!

2. Variation of Parameters Formula

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x), \quad u_1' = -\frac{y_2g(x)}{W(y_1, y_2)}, \quad u_2' = \frac{y_1g(x)}{W(y_1, y_2)}$$

3. Fully Solved Whiteboard Problems

Concordia Exam Classic: Resonance Modification and Coefficient Matching

Problem Statement: Find the general solution of $y'' - 3y' + 2y = 4e^{2x} + 2x$.

- Step 1: Solve Homogeneous Equation

$r^2 - 3r + 2 = 0 \implies (r - 1)(r - 2) = 0 \implies y_c(x) = c_1e^x + c_2e^{2x}.$
- Step 2: Formulate Trial Guess with Resonance Check

For $2x$, guess $y_{p1} = Ax + B$. For $4e^{2x}$, naive guess Ce^{2x} duplicates c_2e^{2x} in y_c ! By the resonance rule, multiply by x : $y_{p2} = Cxe^{2x}$. Complete trial guess: $y_p = Ax + B + Cxe^{2x}$.
- Step 3: Differentiate Trial Guess

$y_p' = A + Ce^{2x} + 2Cxe^{2x}, y_p'' = 4Ce^{2x} + 4Cxe^{2x}.$
- Step 4: Substitute into ODE and Equate Coefficients

$y_p'' - 3y_p' + 2y_p = (4Ce^{2x} + 4Cxe^{2x}) - 3(A + Ce^{2x} + 2Cxe^{2x}) + 2(Ax + B + Cxe^{2x}) = Ce^{2x} + 2Ax + (2B - 3A).$ Equate to $4e^{2x} + 2x$: $C = 4, 2A = 2 \implies A = 1, 2B - 3(1) = 0 \implies B = \frac{3}{2}.$
- Step 5: Assemble General Solution

$y(x) = c_1e^x + c_2e^{2x} + 4xe^{2x} + x + \frac{3}{2}.$

⚠ Common Exam Pitfall & Trapdoor Avoidance

Failure to apply the resonance multiplication rule when the driving function shares roots with the complementary solution is the #1 point-deduction trap on Concordia Test 2 and Final Exams.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

$y(x) = y_c(x) + y_p(x)$. Always compare your trial guess against y_c and multiply by x to resolve duplication.

- Video reference: [Linear Second-Order Differential Equations Part 2: Non-Homogeneous Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.